

Madhura Khandkar

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Education

The Pennsylvania State University

College of Engineering, Bachelor of Science in Computer Science

Sept 2023 – Dec 2026

State College, PA

Technical Skills

Languages: Java, Python, PHP, C++, JavaScript, SQL, HTML/CSS

Frameworks & Tools: FastAPI, React, Laravel, Ruby on Rails, Git, IntelliJ, PyCharm

Cloud: AWS Bedrock, GCP

Libraries: NumPy, Pandas, SciPy, Matplotlib, TensorFlow, Scrapy, Kivy

Experience

Software Engineering Intern

May 2026 – Present

Mile6

Philadelphia, PA

- Eliminated manual grading, saving **6+** hours a week, through a Laravel scoring engine evaluating **140** answer paths across **30+** assessment questions
- Implemented a React/Vite frontend integrated with Laravel **APIs**, enabling dynamic assessment workflows and progress tracking for **10+** clients.
- Streamlined client reporting by automating PDF generation and email delivery through background jobs, saving 3–4 hours of manual effort weekly.

Software Engineering Intern

Oct 2025 – Present

Pennsylvania State University

State College, PA

- Consolidated institutional data processing by developing a Ruby on Rails API that handled **50,000+** educational records across multiple systems.
- Reduced manual processing by **60%** by automating institutional data imports into Activity Insight with Ruby and XML export logic
- Led PR reviews for the API and XML pipeline, catching 3–4 bugs and preventing merge conflicts before release.

Undergraduate Researcher

May 2024 – Dec 2024

Pennsylvania State University

Middletown, PA

- Engineered a motorized 3D-printed throat model that improved biomechanical movement accuracy by **25%**.
- Programmed Python controlled servo systems to simulate anatomically accurate muscle contractions.
- Designed CAD based anatomical models and optimized motor placement to satisfy simulation requirements.

Projects

DigiLib | Python, AWS Bedrock, React, LLaMA 3

April 2026 – Present

- Developed a RAG powered academic research tool that retrieves and ranks credible sources using conversational AI.
- Reduced academic research time by **40%** through dynamic LLM-based retrieval and ranking pipelines.
- Won 2nd place and \$2,000 in the Bardusch Family IdeaMakers Challenge for the tool's sustainable business model.

AI Voice Copilot | Python, JavaScript, React, FastAPI, AWS Bedrock

March 2026 – April 2026

- Created an AI voice copilot providing real-time guidance on web platforms.
- Implemented AWS Bedrock with Anthropic models to deliver recommendations at **800ms** latency.
- Captured user interactions across **3+** sites, streaming 30 clicks per session to **FastAPI** for validation.

Creditwise | React, Firebase, Figma

May 2025 – July 2025

- Generated an web application enabling transfer students to verify credit transfer eligibility across institutions.
- Reduced course search time by **30%** by centralizing credit transfer data into a single student-facing platform.
- Established real time data storage and evaluation logic supporting **30+** course equivalencies across 3 universities.

Leadership / Extracurricular

AWS Cloud Club Outreach Director

Sept 2025 – Present

Pennsylvania State University

- Expanded event participation from 20 to **100+** students by partnering with **20+** technical student organizations.
- Organized 2–3 AWS focused workshops and speaker sessions to expand technical outreach.
- Connected Alumni and Amazon developers to the club, boosting student exposure to new AWS tools.